

Report
Of
BeeSkilled Machine Learning & AI Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 25SCS1003003711
Name of Student: TEJASVI SINGH
Batch: 2025-2029

CANDIDATE'S DECLARATION

I, TEJASVI SINGH do hereby declare that the internship report titled “**BeeSkilled Machine Learning & AI Internship**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name : TEJASVI SINGH

Roll No. : 25SCS1003003711

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **BeeSkilled** for providing me with the opportunity to complete my four-week internship in the field of Artificial Intelligence and Machine Learning.

I am thankful to the trainers and mentors at BeeSkilled for their guidance, support, and valuable feedback throughout the internship. Their practical approach helped me understand and apply various concepts of Machine Learning through assignments and projects

.
I would also like to express my gratitude to IILM University and the faculty members for providing the academic support and encouragement required to complete this internship successfully.

Finally, I am grateful to everyone who supported and encouraged me throughout the internship and helped me successfully complete the three minor projects and one major capstone project.

Thank You.

Signature

Student Name : Tejasvi Singh

Roll No. : 25SCS1003003711

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CERTIFICATE



4. Project Description

4.1 Introduction

The four-week internship at BeeSkilled was completed in the domain of Artificial Intelligence and Machine Learning. The internship provided practical exposure to the concepts and techniques used in machine learning through assignments and hands-on projects.

During the internship, I worked with Python and various machine learning libraries using Anaconda Jupyter Notebook. The work covered data preprocessing, data visualization, supervised learning, unsupervised learning, model training, and model evaluation.

Three minor projects were completed during the internship, including Titanic Survival Data Cleaning, House Price Prediction, and Iris Flower Clustering. A major capstone project on Heart Disease Prediction was also developed using machine learning classification techniques.

The internship helped in understanding the complete machine learning workflow, from preparing datasets and selecting suitable algorithms to training models and evaluating their performance.

4.2 Organization Profile

BeeSkilled is a skill-development and internship platform that provides students with practical, project-based learning opportunities. It focuses on developing industry-relevant technical and problem-solving skills through hands-on training and assignments.

The organization provides learning opportunities in areas such as Artificial Intelligence, Machine Learning, Data Science, and other emerging technologies. As part of the internship program, students work on practical projects to apply their theoretical knowledge and gain experience with relevant tools and technologies.

During my internship, BeeSkilled provided practical exposure to Artificial Intelligence and Machine Learning through assignments, three minor projects, and a major capstone project.

4.3 Problem Statement

The main focus of the internship was to gain practical experience in applying Artificial Intelligence and Machine Learning concepts to real-world datasets. The internship involved working through different stages of the machine learning process, including data collection, data preprocessing, visualization, model building, training, and evaluation.

During the internship, three minor projects were completed to develop practical understanding of data cleaning, regression, and clustering techniques. These projects included **Titanic Survival Prediction – Data Cleaning, House Price Prediction, and Iris Flower Clustering.**

1. Titanic Survival Prediction – Data Cleaning Project

The problem was to clean and prepare the Titanic dataset by handling missing data, encoding categorical variables, and preparing the dataset for further analysis.

2. House Price Prediction Model

The problem was to predict house prices using relevant features and a Linear Regression model, followed by evaluating the model's performance using the R^2 score.

3. Iris Flower Clustering Project

The problem was to group Iris flowers into clusters based on their features using K-Means Clustering and compare the resulting clusters with the actual labels.

4. Heart Disease Prediction – Major Project

The problem was to develop a classification model that predicts whether a patient is likely to have heart disease based on medical attributes, using Logistic Regression and Decision Tree algorithms.

4.4 Project Objectives

The main objectives of the internship were:

- To understand the fundamentals and practical applications of Artificial Intelligence and Machine Learning.
- To gain hands-on experience in data preprocessing, visualization, and analysis.
- To implement supervised and unsupervised machine learning algorithms.
- To develop and evaluate machine learning models using real-world datasets.
- To complete three minor projects and one major Heart Disease Prediction project.
- To improve practical skills in Python, Jupyter Notebook, and machine learning libraries.
- To develop skills in project documentation and presentation.

4.5 Scope of the Project

The scope of this internship was to gain practical experience in applying Artificial Intelligence and Machine Learning techniques to real-world datasets. The work covered the complete machine learning workflow, including:

- Data Preprocessing: Cleaning, handling missing values, encoding, and feature scaling.
- Data Visualization: Exploring and presenting patterns in datasets using graphs and plots.
- Supervised Learning: Applying Linear Regression, Logistic Regression, Decision Tree, and Random Forest algorithms.
- Unsupervised Learning: Applying K-Means Clustering and PCA for data analysis.
- Model Evaluation: Measuring model performance using appropriate evaluation metrics.
- Project Development: Completing three minor projects and one major Heart Disease Prediction project.
- Documentation: Preparing the major project report and presentation.

4.6 Technologies and Tools Used

The following technologies and tools were used during the internship:

Programming Language

- Python

Development Environment

- Anaconda
- Jupyter Notebook

Python Libraries

- Pandas – Data manipulation and preprocessing
- NumPy – Numerical operations
- Matplotlib – Data visualization
- Seaborn – Statistical visualization
- Scikit-learn – Machine learning model development and evaluation

Machine Learning Techniques

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- K-Means Clustering
- Principal Component Analysis (PCA)

4.7 System Architecture

The system architecture followed during the internship consisted of separate workflows for each machine learning project. Each project used a structured process involving dataset preparation, preprocessing, model implementation, and generation of results.

Dataset Collection



Data Preprocessing



Data Exploration & Visualization



Feature Preparation / Scaling



Train-Test Split



Model Training



Model Evaluation



Prediction / Clustering

Different machine learning techniques were applied depending on the project requirements. Regression was used for house price prediction, classification for Titanic and heart disease prediction, and clustering for the Iris project.

4.7.1 Titanic Survival Prediction – Data Cleaning

Titanic Dataset



Data Loading using Pandas



Missing Data Handling



Categorical Encoding



Age Distribution Visualization



Cleaned Dataset → CSV Output

4.7.2 House Price Prediction

House Prices Dataset



Data Preprocessing



Feature Selection & Preparation



Train-Test Split



Linear Regression Model



House Price Prediction



R² Score Evaluation

4.7.3 Iris Flower Clustering

Iris Dataset



Data Preprocessing & Scaling



K-Means Clustering (K=3)



Cluster Visualization



Compare Clusters with True Labels

4.7.4 Heart Disease Prediction

Heart Disease UCI Dataset



Data Cleaning & Encoding



Feature Scaling



Train-Test Split (80:20)



Logistic Regression / Decision Tree



Model Evaluation



Heart Disease Prediction

The overall architecture followed during the internship can be summarized through the following layers:

- Data Layer: Collection and loading of datasets for different machine learning projects.
- Preprocessing Layer: Cleaning data, handling missing values, encoding categorical variables, and scaling features.
- Analysis Layer: Exploring datasets and identifying patterns using statistical analysis and visualization.

- Model Layer: Training appropriate supervised and unsupervised machine learning models based on project requirements.
- Evaluation Layer: Evaluating trained models using suitable performance metrics and visualizations.
- Output Layer: Generating predictions, clusters, visualizations, and processed datasets.
- Documentation Layer: Maintaining Jupyter Notebooks and preparing the major project report and presentation.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on implementation, assignments, and project development. Each week focused on different stages and techniques of Machine Learning, progressing from data preprocessing to model development and finally to a major capstone project.

Week 1 – Machine Learning Fundamentals & Data Preprocessing

Covered Machine Learning basics, dataset exploration, missing-value handling, encoding, scaling, and Train-Test Split. Completed the Titanic Survival Prediction project.

Week 2 – Supervised Learning & Regression

Covered Linear Regression, Logistic Regression, Decision Tree, Random Forest, and model evaluation. Completed the House Price Prediction project.

Week 3 – Unsupervised Learning

Covered K-Means Clustering, PCA, scaling, and the Elbow Method. Completed the Iris Flower Clustering project.

Week 4 – Major Project & Documentation

Developed the Heart Disease Prediction project using the complete Machine Learning workflow, followed by model evaluation, visualization, report preparation, and presentation.

Note: There was no stipend associated with this internship. A final quiz-based assessment was conducted at the end of the internship to evaluate the knowledge gained during the training.

4.9 Expected Outcomes

The internship projects were expected to provide practical understanding and hands-on experience in Artificial Intelligence and Machine Learning. The key expected outcomes were:

- Gain practical knowledge of data preprocessing and visualization.
- Understand and apply supervised and unsupervised learning techniques.
- Develop and evaluate machine learning models using real-world datasets.
- Improve skills in Python and machine learning libraries.
- Understand the complete machine learning workflow from data preparation to prediction.
- Successfully complete three minor projects and one major capstone project.
- Develop a working Heart Disease Prediction model using machine learning techniques.
- Gain experience in project documentation and presentation.

4.10 Certificates of Completion and Communication Proof

The internship completion certificate issued by BeeSkilled is attached as proof of successful completion of the internship program.

Relevant communication emails related to the internship, project submission, and completion of the program are also attached for verification.

Attachments

A. Internship Offer Letter

B. Internship Completion Certificate

C. Communication Mail Proof

A. Internship Offer Letter



INTERNSHIP OFFER LETTER

Date: 30/07/2026

To,
Tejasvi Singh

Subject: Offer For Internship

Dear Tejasvi Singh,

We are pleased to offer you an online internship with **Beeskilled** in the domain of Machine Learning & AI. The internship will commence from 30/07/2026 and end on 10/09/2026.

During this period, you will be required to work on assigned projects and complete the tasks within deadlines. This internship is aimed at enhancing your practical knowledge and hands-on experience.

We look forward to your positive response and hope this internship will contribute meaningfully to your career growth.

Best regards,

Skaur



Simranjeet Kaur
Founder & Program Head

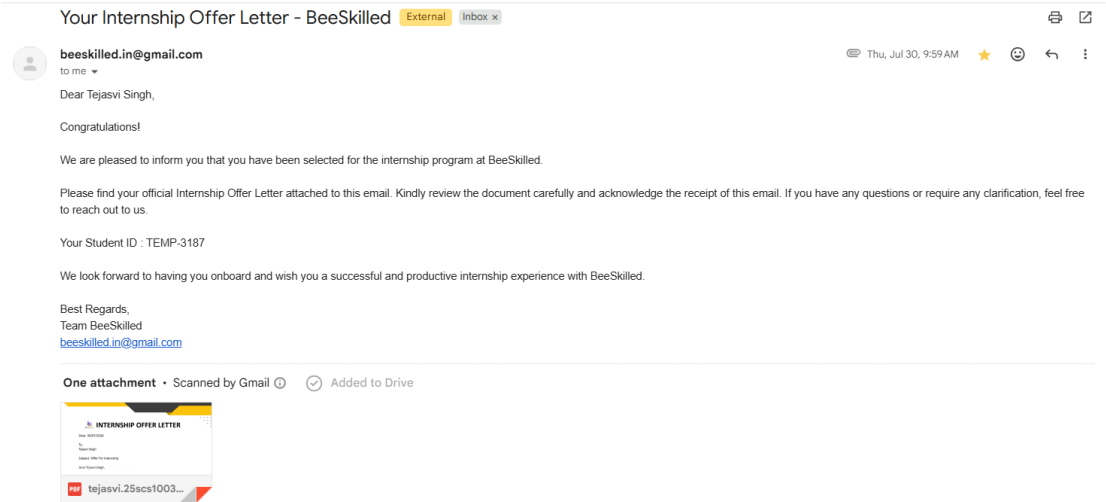


AICTE & MSME REG.

B. Internship Completion Certificate



C. Communication Mail Proof



5. BIBLIOGRAPHY / REFERENCES

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<https://www.kaggle.com/datasets/yasserh/titanic-dataset>
- Kaggle, House Prices Dataset, available at:
<https://www.kaggle.com/datasets/yasserh/housing-prices-dataset>
- UCI Machine Learning Repository, Iris Dataset, available at:
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- Kaggle, Heart Disease UCI Dataset, available at:
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- Scikit-learn Documentation, Machine Learning Algorithms and Evaluation Metrics, available at: <https://scikit-learn.org/>
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- Seaborn Documentation, Statistical Data Visualization, available at: <https://seaborn.pydata.org/>
- BeeSkilled : <https://beeskilled.in/>